Depression of freezing point of the solvent

1. Molal depression constant for water is 1.86^oC. The freezing point of a 0.05 molal solution of a non-electrolyte in water is
(a) -1.86^oC (b) -0.93^oC
(c) -0.093^oC (d) 0.93^oC

2. The amount of urea to be dissolved in 500 *ml* of water (*K* =18.6 *K* $mole^{-1}$in 100*g* solvent) to produce a depression of 0.186^oC in freezing point is
(a) 9 *g* (b) 6 *g*
(c) 3 *g* (d) 0.3 *g*

3. The maximum freezing point falls in
(a) Camphor
(b) Naphthalene
(c) Benzene
(d) Water

4. Which one of the following statements is FALSE
(a) The correct order of osmotic pressure for 0.01 M aqueous solution of each compound is $BaCl_2 > KCl > CH_3COOH >$ sucrose.
(b) The osmotic pressure (π) of a solution is given by the equation $\pi = MRT$ where *M* is the molarity of the solution.
(c) Raoult's law states that the vapour pressure of a component over a solution is proportional to its mole fraction.
(d) Two sucrose solutions of same molality prepared in different solvents will have the same freezing point depression.

5. Solute when dissolved in water
(a) Increases the vapour pressure of water
(b) Decreases the boiling point of water
(c) Decreases the freezing point of water
(d) All of the above

6. The freezing point of a solution prepared from $1.25gm$ of a non-electrolyte and $20gm$ of water is $271.9K$. If molar depression constant is $1.86Kmole^{-1}$, then molar mass of the solute will be
(a) 105.7 (b) 106.7
(c) 115.3 (d) 93.9

7. What is the freezing point of a solution containing $8.1gHBr$ in $100g$ water assuming the acid to be

90% ionised (K_f for water $= 1.86 Kmole^{-1}$)

(a) $0.85^o C$ (b) $-3.53^o C$
(c) $0^o C$ (d) $-0.35^o C$

8. If K_f value of H_2O is 1.86. The value of ΔT_f for $0.1m$ solution of non-volatile solute is
(a) 18.6 (b) 0.186
(c) 1.86 (d) 0.0186

9. 1% solution of $Ca(NO_3)_2$ has freezing point
(a) $0^o C$
(b) Less than $0^o C$
(c) Greater than $0^o C$
(d) None of the above

10. A solution of urea (mol. mass 56*g* *mol*$^{-1}$) boils at 100.18°*C* at the atmospheric pressure. If K_f and K_b for water are 1.86 and 0.512*K kg mol*$^{-1}$ respectively the above solution will freeze at
(a) – 6.54°*C* (b) 6.54°*C*
(c) 0.654°*C* (d) –0.654°*C*

11. The molar freezing point constant for water is $1.86^o C mole^{-1}$. If 342 gm of canesugar $(C_{12}H_{22}O_{11})$ are dissolved in 1000 gm of water, the solution will freeze at
(a) $-1.86^o C$ (b) $1.86^o C$
(c) $-3.92^o C$ (d) $2.42^o C$

12. An aqueous solution of a non-electrolyte boils at $100.52^o C$. The freezing point of the solution will be
(a) $0^o C$
(b) $-1.86^o C$
(c) $1.86^o C$
(d) None of the above

13. The freezing point of one molal $NaCl$ solution assuming $NaCl$ to be 100% dissociated in water is (molal depression constant = 1.86)
(a) $-1.86^o C$ (b) $-3.72^o C$
(c) $+1.86^o C$ (d) $+3.72^o C$

14. Heavy water freezes at
(a) $0^o C$ (b) $3.8^o C$
(c) $38^o C$ (d) $-0.38^o C$

15. After adding a solute freezing point of solution decreases to – 0.186. Calculate ΔT_b if $K_f = 1.86$ and $K_b = 0.521$.
(a) 0.521 (b) 0.0521
(c) 1.86 (d) 0.0186

16. Given that ΔT_f is the depression in freezing point of the solvent in a solution of a non-volatile solute of molality m, the quantity $\lim_{m \to 0} \left(\frac{\Delta T_f}{m}\right)$ is equal to

(a) Zero
(b) One
(c) Three
(d) None of the above

17. The freezing point of 1 percent solution of lead nitrate in water will be
(a) Below 0^oC (b) 0^oC
(c) 1^oC (d) 2^oC

18. What is the effect of the addition of sugar on the boiling and freezing points of water
(a) Both boiling point and freezing point increases
(b) Both boiling point and freezing point decreases
(c) Boiling point increases and freezing point decreases
(d) Boiling point decreases and freezing point increases

19. During depression of freezing point in a solution the following are in equilibrium
(a) Liquid solvent, solid solvent
(b) Liquid solvent, solid solute
(c) Liquid solute, solid solute
(d) Liquid solute solid solvent

20. 1.00 *gm* of a non-electrolyte solute dissolved in 50 *gm* of benzene lowered the freezing point of benzene by 0.40 *K*. K_f for benzene is 5.12 *kg mol*$^{-1}$. Molecular mass of the solute will be
(a) $256 gmol^{-1}$
(b) $2.56 gmol^{-1}$
(c) $512 \times 10^3 gmol^{-1}$
(d) $2.56 \times 10^4 gmol^{-1}$

21. 0.440 *g* of a substance dissolved in 22.2 *g* of benzene lowered the freezing point of benzene by 0.567^oC. The molecular mass of the substance $(K_f = 5.12^oCmol^{-1})$
(a) 178.9 (b) 177.8
(c) 176.7 (d) 175.6

22. Which of the following aqueous molal solution have highest freezing point
(a) Urea
(b) Barium chloride
(c) Potassium bromide
(d) Aluminium sulphate

23. Which will show maximum depression in freezing point when concentration is 0.1*M*
(a) *NaCl* (b) Urea
(c) Glucose (d) K_2SO_4

24. The freezing point of a 0.01*M* aqueous glucose solution at 1

atmosphere is -0.18^oC. To it, an addition of equal volume of 0.002 *M* glucose solution will; produce a solution with freezing point of nearly

(a) -0.036^oC (b) -0.108^oC
(c) -0.216^oC (d) -0.422^oC

25. What should be the freezing point of aqueous solution containing $17gm$ of C_2H_5OH in $1000gm$ of water (water K_f = 1.86 $deg - kgmol^{-1}$

(a) -0.69^oC (b) -0.34^oC
(c) 0.0^oC (d) 0.34^oC

26. In the depression of freezing point experiment, it is found that the

(a) Vapour pressure of the solution is less than that of pure solvent
(b) Vapour pressure of the solution is more than that of pure solvent
(c) Only solute molecules solidify at the freezing point
(d) Only solvent molecules solidify at the freezing point

27. Calculate the molal depression constant of a solvent which has freezing point 16.6^oC and latent heat of fusion $180.75Jg^{-1}$.

(a) 2.68 (b) 3.86
(c) 4.68 (d) 2.86t6